

Opeongo Computational Biology Home Assignment

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Key Findings:

- The ECM panel behaves as a stromal program: fibroblast-lineage cells express all ten genes, pericytes a collagen subset, and endothelial cells mainly SPARC and FN1.
- In endothelium (the only population with paired coverage in all cohorts), SPARC was higher in tumor, robust to removing mesenchymal marker-positive cells but cohort-dependent: driven by B, null in A, reversed in C.
- In cohort A's linked bulk RNA and protein, SPARC and FN1 were higher in tumor in every paired donor: directionally consistent, but the increase cannot be attributed to endothelial source specifically.

1 Quality assessment and cohort integration

Contents. A (GSE227898; lead cohort, the only one with linked bulk RNA and proteomics) is single-nucleus RNA: 34 specimens from 25 donors, 141,950 author-annotated nuclei identified in the raw droplet matrices. B (GSE178481) is single-cell RNA: 26 specimens, 157,881 annotated cells with author QC fields. C (GSE159115, supplied exercise derivative) is single-cell RNA: 13 ccRCC/benign-adjacent specimens from 8 donors, 29,499 cells, no cell labels; one donor contributes cortex and medulla, which were combined as that donor's adjacent sample. Features were matched by Ensembl ID (A, C) and by symbol (B); duplicated symbols (A 24, C 34) were summed, and absent features were never treated as zeros. All ten panel genes occur in all cohorts.

QC criteria. nCount, nFeature and %MT were recomputed per specimen from the author/deposited cell sets (B's values matched its supplied TotalUMI, gene.num and MT.ratio exactly, $r = 1.000$), and thresholds were set per cohort rather than shared across assays (Table 1, Fig. 1). A was already author-filtered on depth (1,000–10,000 UMIs, $\leq 10\%$ MT), so all of its removal (12.6%) came from the 2% nuclear MT cap. B's author floor (~700 UMIs) lies below our depth thresholds, so most of its loss (18.9%) came from depth/complexity, mainly small lymphocytes, while the 10% MT cap removed mitochondria-rich epithelium (4.6%). C has a ~300-gene ceiling and broad %MT, reported as properties of the derivative rather than pass/fail diagnostics; because tubule-rich kidney cells are mitochondria-rich and C's median non-MT UMIs stayed ≥ 200 up to 95% MT, C was filtered on analyzability (≥ 150 non-MT UMIs per cell) plus a 95% MT cap rather than a fixed low cutoff. B's supplied doublet scores were inspected but not thresholded; the nFeature <4,500 cap is the only doublet guard, and no total-count ceiling was applied to C, whose high-count cells are MT-dominated rather than doublet-like. No specimen was excluded, including two MT-dominated C tumor specimens (GSM4819728, GSM4819726; 55% and 36% of cells retained). **Observation:** A's MT cap removed immune nuclei disproportionately (13.9% of adjacent lymphoid retained); at 5%, retention rose to 94.5% overall and 48.6% for adjacent lymphoid, so A's immune composition is not interpreted. Endothelial paired coverage (≥ 20 cells) was unchanged by QC in A and B and identical at 2% and 5% MT.

Cohort	Assay	Specimens	Num. Initial Cells	Num. Cells after QC	%Retained Cells	Cell-level criteria (specimens kept if ≥ 20 cells)
A	snRNA	34 (25 donors; 4 adjacent)	141,950	124,088	87.4%	nFeature 500 - 4,500; nCount >1,000; %MT <2%
B	scRNA	26 (11 donors; 10 adjacent)	157,881	120,752	76.5%	nFeature 500 - 4,500; nCount >1,000; %MT <10%
C	scRNA (derivative)	13 (8 donors; 6 adjacent)	29,499	22,191	75.2%	nFeature 100 - 400; non-MT UMIs ≥ 150 ; %MT <95%

Table 1. Cohort contents and cell-level QC. No specimens were removed. Smallest retained specimen: A: 1,138, B: 759, C: 409 cells. Per-specimen and per-population QC tables are listed in the repository README.

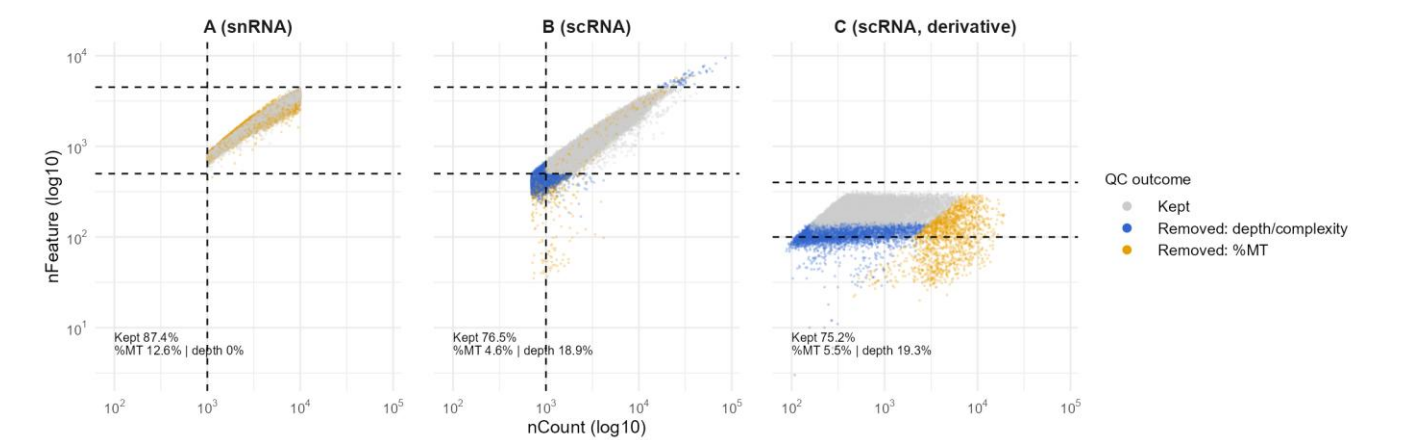


Figure 1. Cell-level QC outcome per cohort: unfiltered cells, nCount vs nFeature (log10), colored by outcome; dashed lines mark depth (nCount, UMIs per cell) and complexity (nFeature, genes per cell) thresholds, and panel labels give percentages. Cohort A was author-filtered on depth, so all its removal came from the 2% MT cap; B's author floor (~700 UMIs) lies below the applied depth thresholds; C's removals mainly reflect the ≥ 150 non-MT UMIs per cell criterion, which depends on %MT rather than a plotted axis, so its boundary appears smeared instead of following a dashed line.

Integration. 2,000 integration features were selected jointly across A, B and C (all present in each cohort), each cohort was scaled and reduced separately, and cohorts were integrated as equals with Seurat RPCA anchors [2-3] (30 dimensions; k.weight 30 for memory). UMAP and clustering used 10 PCs, where both elbow plots flatten. The integrated embedding was used only for visualization and labelling, and not for expression testing. **Unintegrated baseline (Fig. 2a):** cells separate by cohort; A forms its own islands and C bridges toward B, consistent with assay type (nuclei vs whole cells) dominating the structure. **RPCA (Fig. 2b-c):** A and B cells of the same population co-localize in shared clusters (immune, endothelial, pericyte), so identity is preserved and the cohort effect removed. At resolution=1, 33 of 37 clusters were $\geq 80\%$ pure for a single A/B population, with residual mixing at biological boundaries (pericyte/fibroblast, tumor/normal epithelium). Malignant epithelium remains partly A-dominated (nuclei-specific capture or between-cohort tumor heterogeneity). C mixes only partly: it reaches the endothelial cluster, but some C cells lie on bridges between populations, consistent with low per-cell information.

Labels for C. A/B populations come from the supplied harmonized guide (Unresolved retained in coverage, excluded from contrasts). Each C cell received the majority A/B label of its resolution=1 cluster when that label covered $\geq 80\%$ of ≥ 50 A/B cells; 32/37 clusters passed, leaving 10.8% of C unresolved, and no C cell reached Stromal_fibro_myo. Labels were checked with canonical markers outside the panel: 11/13 peaked in the expected population (CDH5/PECAM1 endothelial, CD68/LYZ myeloid, CD3D lymphoid, CA9 malignant, EPCAM nonmalignant epithelial, RGS5/PDGFRB pericyte, PDGFRA fibroblast); NDUFA4L2 and ACTA2 peaked in pericytes. Unresolved clusters co-expressed markers of two lineages, consistent with doublets or mixed states. C labels are broad and method-sensitive. **Decision:** all three cohorts are included for cell identity and coverage; C's quantitative effects are assessed separately (Section 2). **What combining adds:** A is mostly tumor-only (2 paired endothelial donors), B contributes most tumor-adjacent pairs and the only explicit pericyte label, and C provides a third cohort to test replication.

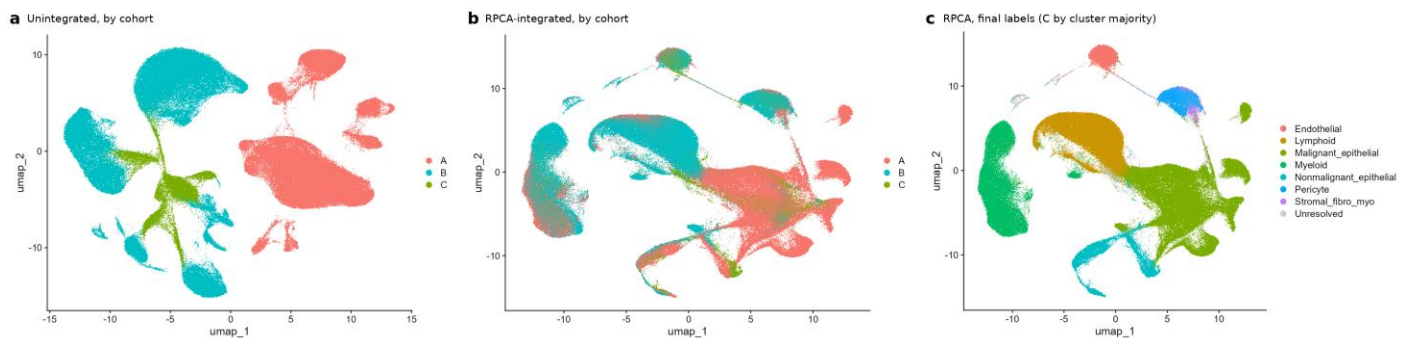


Figure 2. (a) Unintegrated A+B+C UMAP by cohort. (b) RPCA-integrated UMAP by cohort. (c) Final labels: A/B from the harmonized author annotations; C by resolution=1 cluster majority.

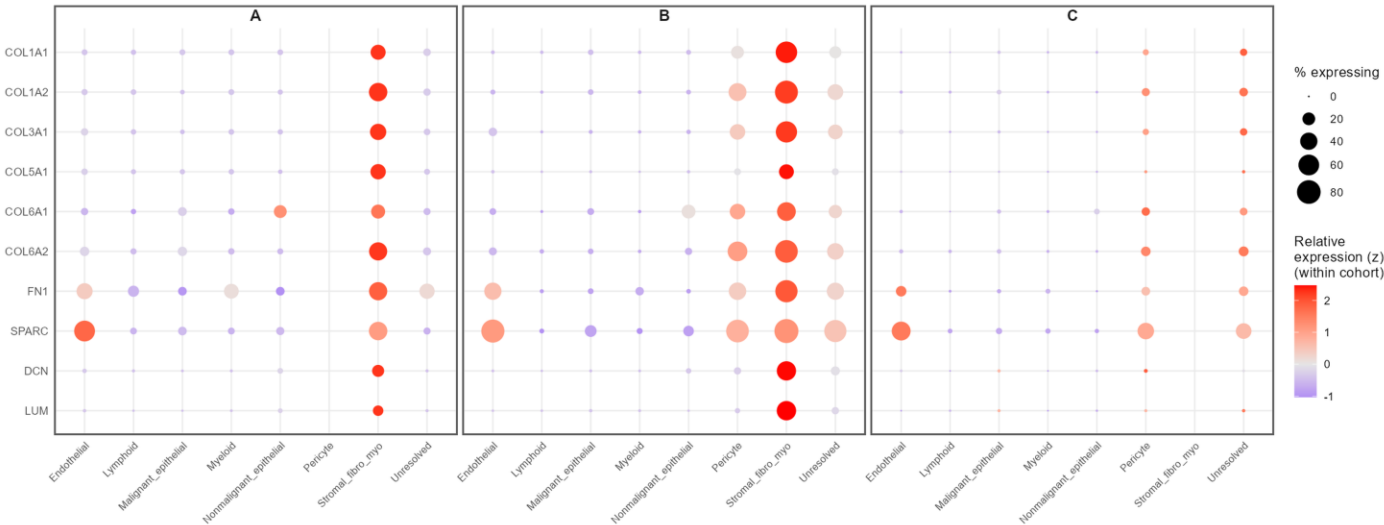
2 The ECM program

Overview (Fig. 3a). Observation: Stromal_fibro_myo is highest for 7/10 genes, and essentially the only source of DCN, LUM and COL5A1; pericytes peak for COL6A1/COL6A2 and express a collagen subset; endothelial cells express SPARC (highest population, 64.5% of cells combined across cohorts) and FN1; immune and epithelial cells are near silent. The same population ranking holds within each cohort; C is sparser, and differences are visible: no pericyte annotations in A and no Stromal_fibro_myo cells in C. Combined cell-level views by condition (repository) suggest gene-specific tumor shifts (e.g. lower DCN/LUM with higher FN1 in fibroblasts). **Interpretation:** the panel is not one co-regulated program across cell types, so effects are reported per gene alongside a single summary.

Population choice. Donors with ≥ 20 cells in both tumor and adjacent tissue: Endothelial=13 (A=2, B=8, C=3); Pericyte=11 (B=9, C=2; no pericyte label in A); Stromal_fibro_myo=2 (A only; 4 at ≥ 10 cells; no C cells; Fig. 3a). Beyond coverage, fibroblast-lineage labels are not harmonized (A combines fibroblasts and myofibroblasts, and B's fibroblasts and pericytes are not established equivalents), so tumor-adjacent differences there would mix identity, composition and expression. Endothelial cells are the only candidate with consistent identity and paired coverage in all three cohorts. Examining matrix genes in endothelial population is also biologically relevant: a previous study shows that ccRCC tumor endothelial cells exhibit increased ECM-organization pathways relative to matched normal endothelium [1]. The endothelial finding therefore concerns the part of the panel this population expresses, SPARC and FN1. Malignant epithelium (no adjacent counterpart) and B nonmalignant epithelium (absent from tumor specimens) are non-estimable.

Testing. Raw UMI counts from the RNA assay (not batch-corrected) were summed per donor $\times$ condition within endothelial cells, combining all of a donor's specimens for that condition into one sample (pseudobulk; 48 donor $\times$ condition samples with ≥ 20 cells, from 13 paired and 22 single-condition donors). Donors are the unit of replication. Libraries were TMM-normalized (edgeR) and modeled with limma-voom [4-5] on all expressed genes, with cohort as a covariate and donor as a blocking factor (duplicateCorrelation), which uses pairing where available and keeps unpaired donors; a strictly paired model ($\sim$ donor + condition) was fitted on the 13 paired donors. The summary is the panel-sum log2FC (summed counts of all ten genes per sample), defined identically in every model and, in endothelium, dominated by SPARC and FN1. Collagens, DCN and LUM mostly failed expression filtering in endothelium and are reported as not estimable, not unchanged. Plots use log-normalized RNA values.

a ECM panel by population, per cohort (z-scored within cohort)



b Endothelial pseudobulk, 13 paired donors (logCPM)

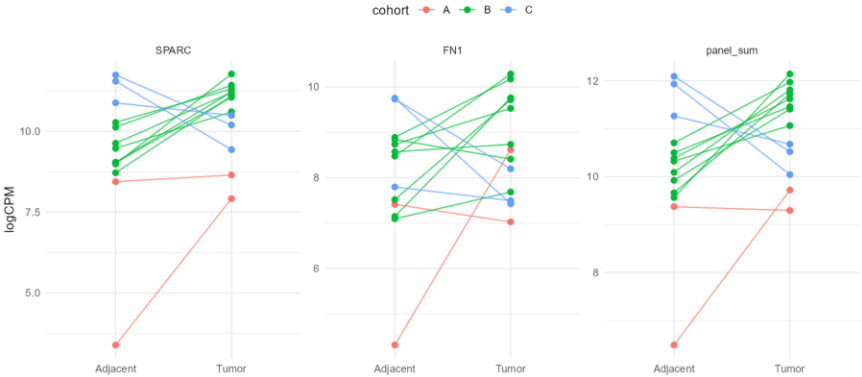


Figure 3. (a) ECM-panel expression across major populations, shown separately for cohorts A, B and C (color = gene-wise z-score of mean log-normalized expression within each cohort; size = % expressing). Colors and dot sizes compare populations within a cohort only; absolute levels are not comparable across cohorts because of assay differences (nuclei, whole cells, derivative counts). (b) Endothelial pseudobulk Adjacent to Tumor for SPARC, FN1 and the panel sum in the 13 paired donors, colored by cohort.

Model	Donors (paired +unpaired)	Panel-sum log2FC (95% CI), P-val.	SPARC log2FC (95% CI)	FN1 log2FC (95% CI)	Panel-sum log2FC after Marker-cleaning (95% CI)
A only	17 (2 + 15)	+0.11 (-0.92, 1.13), 0.83	-0.08 (-1.21, 1.04)	-0.23 (-1.48, 1.03)	-0.14 (-1.06, 0.77)
B only	11 (8 + 3)	+1.62 (1.24, 2.01), 9.3×10 ⁻⁸	+1.84 (1.47, 2.20)	+1.27 (0.60, 1.94)	+1.63 (1.24, 2.02)
C only	7 (3 + 4)	+0.06 (-1.60, 1.72), 0.94	-0.25 (-1.50, 1.01)	-0.79 (-1.78, 0.19)	+0.03 (-1.61, 1.66)
A+B	28 (10 + 18)	+1.23 (0.71, 1.75), 3.0×10 ⁻⁵	+1.27 (0.64, 1.90)	+0.86 (0.15, 1.56)	+1.17 (0.66, 1.67)
A+B+C	35 (13 + 22)	+0.70 (0.12, 1.27), 0.018	+0.88 (0.24, 1.53)	+0.36 (-0.28, 1.00)	+0.68 (0.09, 1.27)
Paired only, A+B+C	13 (13 + 0)	+0.86 (-0.06, 1.78), 0.065	+1.04 (0.22, 1.85)	+0.64 (-0.18, 1.46)	+0.84 (-0.06, 1.73)

Table 2. Endothelial tumor-vs-adjacent effects as cohorts are added, before and after removing mesenchymal marker-positive cells. Positive = higher in tumor. After marker cleaning, SPARC was unchanged (B +1.89, A+B +1.27, A+B+C +0.86, paired +1.07) while FN1 weakened (B +1.09; A+B +0.62, A+B+C +0.16). Cohort × condition interaction (A+B+C): SPARC P = 0.016, FN1 P = 0.026, COL3A1 P = 0.025 (BH-FDR 0.12-0.17).

Effect, donor variability and adding cohorts. Observation: in B, SPARC was higher in tumor in all 8 paired donors and FN1 in 7 of 8 (Fig. 3b), giving a precise panel-sum increase (+1.62, ≈3-fold). A alone shows no detectable effect: its two paired donors split, with the apparent increase in C3N-01200 resting on an adjacent sample of only 26 cells. In C, all three paired donors were lower in tumor, driven by high adjacent levels; with four unpaired donors included, C-only estimates are negative for SPARC (-0.25) and FN1 (-0.79) but with wide intervals that include zero. In the strictly paired model across cohorts, SPARC remained higher in tumor (+1.04, 95% CI 0.22-1.85; unadjusted P=0.015), whereas the panel sum did not reach significance. Adding B to A produces the signal; adding C reduces it (1.23 to 0.70). The interaction is nominally significant for SPARC and FN1 but does not survive FDR correction with three paired C donors. SPARC, FN1 and panel-sum differences are consistent within B (all positive) and within C (all three negative), while A's two donors split, so donor-level variability is mainly between cohorts. Collagens, DCN and LUM vary in sign or are undetected (per-gene plot in repository), so the panel-sum result is effectively a SPARC/FN1 result. **Interpretation:** the tumor-associated endothelial increase is supported mainly by B and is not replicated across cohorts. C's discordance coincides with its derivative counts, and C's paired tumor specimens include the

two MT-dominated ones (GSM4819728, GSM4819726; median %MT 92% and 96%), so its tumor-side endothelial profiles may rest on low-information cells. The reversal may therefore reflect the supplied derivative rather than biology, but this was not tested.

Robustness check (main uncertainty: is the signal endothelial?). SPARC and FN1 are highly expressed by pericytes and fibroblasts, which neighbor endothelium in the embedding, so mesenchymal cells (pericytes or fibroblasts) within the endothelial label could create the tumor increase. Endothelial-labeled cells with any count of RGS5, PDGFRB, ACTA2, PDGFRA, MYH11 or NOTCH3 (mesenchymal markers outside the panel, used only where present in all three cohorts) were removed and all models refitted unchanged. Flagged fractions were balanced in A (19.3% tumor vs 19.2% adjacent) and C (9.0% vs 8.1%) but higher in B tumor (32.4% tumor vs 9.2% adjacent); even in B, removing flagged cells left the SPARC and panel-sum effects essentially unchanged (Table 2), arguing against contamination as the source of the SPARC signal. However, FN1 was partly sensitive (A+B: +0.86, P = 0.019 changes to +0.62, P = 0.093). **Conclusion:** the tumor-associated increase in endothelial SPARC is robust to removing mesenchymal marker-positive cells; FN1 is weaker and contamination-sensitive; C's discordance is not explained by this mechanism. The check does not address ambient (cell-free) RNA within retained cells or contaminants with undetected markers.

3 Linked modalities (cohort A)

Correspondence and matched subset. The crosswalk covers cohort A only (34 snRNA specimens, 25 donors); snRNA and protein are linked by author-reconciled parent specimen; bulk RNA is linked by donor and condition only, without aliquot reconciliation, so assays use different portions of the same parent tissue. 27/34 snRNA specimens (24 donors; 23 tumor, 4 adjacent) have an unambiguous parent-specimen protein match plus a donor/condition bulk RNA match. Exclusions: five snRNA specimens without an author bulk link are extra tumor aliquots of donors already matched (C3L-00088, C3L-00416, C3N-00733, C3N-01200 ×2; no coverage lost); C3L-00908 tumor (protein from a different parent specimen) and C3N-00242 tumor (protein maps to two parent specimens) were excluded from the protein comparison, while C3N-00242 remains usable for bulk RNA. Supplied bulk values (UQ-normalized log2 RNA; log2 reference-normalized protein) were used as provided, without re-normalization or imputation; all ten panel genes have finite values in the matched columns. Tumor-adjacent differences were computed within donor in each assay: 4 paired donors for bulk RNA and 3 for protein.

Donor	Endothelial snRNA cells (Tumor / Adj)	snRNA endothelial SPARC / FN1	Bulk RNA SPARC / FN1	Protein SPARC / FN1
C3L-00079	23 / 261	+0.21 / -0.39	+2.07 / +3.51	+1.78 / +0.91
C3N-01200	173 / 26	+4.52 / +4.30	+2.02 / +5.17	+0.77 / +1.37
C3L-00088	556 / 0	not estimable (0 adjacent cells)	+1.51 / +1.00	+0.52 / +0.38
C3N-00242	<20 / 104	not estimable (<20 tumor cells)	+0.89 / +0.49	excluded (protein maps to 2 parent specimens)
All paired donors	2 donors with paired snRNA	SPARC 2/2 positive; FN1 1/2	4/4 positive for both; mean +1.62 / +2.54	3/3 positive for both; mean +1.03 / +0.89

Table 3. Within-donor tumor – adjacent differences (log2 scale). The first two donors have paired endothelial snRNA data (≥20 cells per side) and form the primary comparison. With 3–4 donors, p-values are descriptive only; the evidence is the direction.

Agreement. SPARC and FN1 were higher in tumor in every paired donor in both whole-tissue assays. The two donors with paired endothelial snRNA data, the most directly comparable subset, show the same direction in bulk RNA and protein. **Discrepancies.** In C3L-00079, bulk RNA and protein rise clearly while endothelial snRNA is flat, so the tissue-level increase may not come from endothelium; in C3N-01200 all three assays rise, but the snRNA estimate rests on 26 adjacent cells. The unweighted mean of all ten genes showed no consistent bulk direction (RNA -0.42, protein +0.08), because whole-tissue collagens, DCN and LUM track stromal content; it is not equivalent to the endothelial panel sum. Cohort A's own snRNA endothelial effect was null, so this comparison shows that the direction holds in A's tissue, not an independent replication of the B-driven cell-level effect. **Limitations.** *Composition:* ccRCC is highly vascular and stroma-rich relative to tubule-dominated adjacent kidney, so higher tissue SPARC and FN1 may reflect more endothelial and stromal cells rather than higher per-cell expression. *Cellular source:* single-cell RNA identifies which annotated cells express these genes, not where bulk protein comes from; tissue protein also includes plasma FN1, platelet SPARC and matrix-deposited protein from any cell type. *Coverage and missingness:* one cohort, 4 and 3 paired donors, only 2 with endothelial snRNA on both sides; adjacent specimens are scarce, and bulk RNA linkage is at donor/condition level only.

Code and data availability: https://github.com/Ronen-DS/ECM_kidney_single_cell

References

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